

DD_FUL-08 - Anchor Date Change - Developer Implementation Guide

Version: v1.0 Bundle: 06_fulfillment Companion DD: DD_FUL-08-Anchor_Date_Change-v1.0

1. Goal

Implement the flow that changes a subscription billing anchor date without breaking billing cycle integrity.

FUL-08 coordinates:

- subscription read/update through SUB-LM-01;
- billing preview/apply through BIL APIs;
- configurable approval through EM-CFG-04;
- optional Camunda workflow for async approval/payment/scheduled apply.

2. Runtime Flow

```
sequenceDiagram
    participant B0 as Backoffice/Self-care
    participant FUL as FUL-08
    participant SUB as SUB-LM-01
    participant BIL as BIL-01/BIL-02
    participant APR as EM-CFG-04

    B0->>FUL: POST preview
    FUL->>SUB: read subscription
    FUL->>BIL: preview financial impact
    B0->>FUL: POST create request
    FUL->>APR: evaluate/request approval if required
    APR-->>FUL: approval callback
    FUL->>SUB: apply anchor update
    FUL->>BIL: apply fee/proration
    FUL-->>B0: request APPLIED
```

3. Step 1 - Migrations

Create:

- anchor_date_change_request
- anchor_date_change_preview
- anchor_date_change_apply_log

Add:

- unique active-request guard on (operator_code, subscription_id) for statuses DRAFT, VALIDATING, PENDING_APPROVAL, PENDING_PAYMENT, APPROVED;
- index on approval_request_id;
- index on request status/date.

4. Step 2 - Preview Endpoint

Endpoint:

```
POST /api/subscriptions/SUB-700045/anchor-date-change/preview
Content-Type: application/json
```

Request:

```
{
  "operatorCode": "WIK",
```

```

    "requestedAnchorDay": 15,
    "effectiveMode": "NEXT_CYCLE",
    "reasonCode": "CUSTOMER_REQUEST",
    "sourceChannel": "BACKOFFICE"
  }

```

Flow:

1. Validate day between 1 and 28.
2. Load subscription via SUB-LM-01.
3. Reject if subscription state is not allowed.
4. Check no pending anchor change exists.
5. Build Drools fact and run eligibility.
6. Call Billing preview:

```

POST /api/billing/anchor-date-change/preview
Content-Type: application/json

```

```

{
  "operatorCode": "WIK",
  "subscriptionId": "SUB-700045",
  "currentAnchorDay": 1,
  "requestedAnchorDay": 15,
  "effectiveMode": "NEXT_CYCLE"
}

```

7. Return preview to caller.

Preview does not update subscription or billing ledger.

5. Step 3 - Create Request

Endpoint:

```

POST /api/subscriptions/SUB-700045/anchor-date-change
Idempotency-Key: adc-sub-700045-202607-anchor15
Content-Type: application/json

```

Flow:

1. Validate idempotency key.
2. Re-run validation; do not trust old UI state.
3. Store anchor_date_change_request.
4. Store anchor_date_change_preview.
5. Run anchor-date-change-approval.drl.
6. If approval required, call EM-CFG-04.
7. If payment required, set PENDING_PAYMENT.
8. Otherwise set APPROVED.
9. Emit AnchorDateChangeRequested.

6. Step 4 - EM-CFG-04 Approval

Evaluate:

```

{
  "operatorCode": "WIK",
  "moduleCode": "FUL-08",
  "actionCode": "ANCHOR_DATE_CHANGE_APPROVAL",
  "subjectType": "ANCHOR_DATE_CHANGE",
  "subjectId": "adc-001",
  "context": {
    "subscriptionId": "SUB-700045",
    "currentAnchorDay": 1,
    "requestedAnchorDay": 15,
    "prorationAmount": 1400.00,

```

```

    "currencyCode": "KES"
  }
}

```

If required, start request:

```

POST /api/approvals/requests
Idempotency-Key: approval-anchor-adc-001

```

Store:

- approval_request_id;
- approval_mode;
- camunda_process_instance_id if EM-CFG-04 returns one.

Camunda note: FUL-08 may have its own Camunda process for anchor-change orchestration. EM-CFG-04 may also use Camunda internally for approval only if the approval policy is BPMN. Do not mix the two process responsibilities.

7. Step 5 - Approval Callback

Endpoint:

```

POST /internal/anchor-date-change/adc-001/approval-outcome
Content-Type: application/json

{
  "approvalRequestId": "apr-adc-001",
  "outcome": "APPROVED",
  "decisionComment": "Approved customer billing date alignment."
}

```

Flow:

1. Load request.
2. Verify approval request id.
3. If already processed, return 200.
4. If rejected, set REJECTED and emit event.
5. If approved and payment required, set PENDING_PAYMENT.
6. If approved and no payment required, set APPROVED.

8. Step 6 - Apply

Endpoint:

```

POST /api/anchor-date-change/adc-001/apply
Idempotency-Key: apply-adc-001

```

Flow:

1. Load request and preview.
2. Validate status APPROVED or payment-cleared PENDING_PAYMENT.
3. Insert anchor_date_change_apply_log attempt.
4. Call SUB-LM-01:

```

POST /api/subscriptions/SUB-700045/anchor-date
Idempotency-Key: sub-anchor-adc-001
Content-Type: application/json

{
  "operatorCode": "WIK",
  "newAnchorDay": 15,
  "effectiveAt": "2026-07-15T00:00:00Z",
  "originRefType": "ANCHOR_DATE_CHANGE",
  "originRefId": "adc-001"
}

```

5. Call Billing apply if preview has financial impact:

```

POST /api/billing/anchor-date-change/apply
Idempotency-Key: bil-anchor-adc-001
Content-Type: application/json

{
  "operatorCode": "WIK",
  "subscriptionId": "SUB-700045",
  "requestId": "adc-001",
  "billingPreviewRef": "bil-prev-adc-001"
}

```

6. Mark request APPLIED.

7. Emit AnchorDateChanged.

If Billing fails after SUB update, retry the billing apply with the same idempotency key. Do not create a second SUB update.

9. Drools Samples

```

package com.sophix.fulfillment.anchor

rule "Self care anchor change not allowed for dunning subscriptions"
when
  $f : AnchorChangeFact(sourceChannel == "SELF_CARE", dunningState != "CLEAR")
  $d : AnchorChangeDecision()
then
  $d.reject("DUNNING_SELF_CARE_NOT_ALLOWED");
end

rule "Large proration requires supervisor approval"
when
  $f : AnchorChangeFact(prorationAmount >= 1000.00, currencyCode == "KES")
  $d : AnchorChangeDecision()
then
  $d.setApprovalRequired(true);
  $d.setApprovalPolicyCode("ANCHOR_CHANGE_SUPERVISOR");
end

```

10. Worker Jobs

Job	Purpose
anchor-change-apply-due	Applies scheduled approved requests.
anchor-change-payment-watch	Correlates payment applied events to pending requests.
anchor-change-apply-retry	Retries failed SUB/Billing apply calls.
anchor-change-outbox-publish	Publishes events.

These can run in the shared fulfillment worker deployment.

11. Tests

- Preview rejects invalid day 29.
- Preview calls Billing and returns proration.
- Create request stores preview.
- Approval required request starts EM-CFG-04.
- Approval callback is idempotent.
- Apply calls SUB-LM-01 once.
- Apply calls Billing once with idempotency.
- Retry after Billing failure does not repeat SUB update.
- BIL-03 uses new anchor only after applied event/state.